

Eco Design MCQ Exam

1. What is the main goal of an eco-audit in product design?

- A. To guide decision-making with sufficient precision regarding environmental impacts
- B. To maximize product performance
- C. To increase the aesthetic value of the product
- D. To reduce product cost

Answer: A

2. Which of the following is a correct phase included in the Life Cycle Assessment (LCA)?

- A. Packaging design
- B. Advertisement strategy
- C. End of life
- D. Concept ideation

Answer: C

3. What is the ISO standard related to Life Cycle Assessment?

- A. ISO 9001
- B. ISO 14040
- C. ISO 45001
- D. ISO 26000

Answer: B

4. In the context of eco-design, which strategy minimizes CO₂ emissions during transport?

- A. Choosing the longest route
- B. Using non-recyclable packaging
- C. Increasing transport frequency
- D. Minimizing distance and optimizing transport type

Answer: D

5. What is one key action to reduce the impact during the 'Material' phase of a product's life cycle?

- A. Use materials with the highest strength only
- B. Increase product mass
- C. Maximize thermal losses
- D. Select materials with the lowest embodied energy and CO2 footprint

Answer: D

6. According to the eco-audit for the jug kettle, which phase contributes the most to life energy?

- A. End of life
- B. Use
- C. Transport
- D. Manufacture

Answer: B

7. Which component of the CES EduPack eco-audit tool does not require user input?

- A. Bill of materials
- B. CO2 footprints
- C. Duty cycle
- D. Transport needs

Answer: B

8. What is a recommended end-of-life option in eco-design?

- A. Refurbishment
- B. Combustion without filtration
- C. Exporting to landfill
- D. Incineration with plastics

Answer: A

9. Which of the following is true about full LCA?

- A. Expensive and detailed
- B. Always precise
- C. Cheap and quick to conduct
- D. Simple and standardized

Answer: A

10. What is used to roll up the environmental impact into a single value?

- A. LCA score
- B. Eco-indicator
- C. Carbon index
- D. Impact inventory

Answer: B

11. What can help minimize thermal and electrical losses in the Use phase?

- A. Efficient insulation and design
- B. Adding redundant systems
- C. Increasing mass
- D. Using low-efficiency components

Answer: A

12. Which of the following transport methods has the highest CO₂ emissions per tonne.km?

- A. Sea freight
- B. 32 tonne truck
- C. Air freight
- D. Rail transport

Answer: C

13. Which factor is considered in the Use phase of an eco-audit?

- A. Manufacturing cost
- B. Energy consumption during use
- C. Sales price
- D. Design aesthetics

Answer: B

14. Which process is considered in the 'Material and process energy / CO₂' step?

- A. Market analysis
- B. Design simulation
- C. Casting and molding
- D. Product advertisement

Answer: C

15. In an eco-design strategy, what follows the analysis of results?

- A. Sketching
- B. Prioritization and action recommendation
- C. Prototyping
- D. Marketing

Answer: B

16. What is a key benefit of fast eco-audits?

- A. Lower production costs
- B. Quick decision support
- C. Unregulated product design
- D. Immediate manufacturing

Answer: B

17. Which of the following is a valid end-of-life treatment option?

- A. Weight scaling
- B. Reuse
- C. Color coating
- D. Polishing

Answer: B

18. What type of database does the CES EduPack tool include?

- A. Embodied energies and CO2 data
- B. Customer preferences
- C. Manufacturing delays
- D. Color templates

Answer: A

19. Which energy source is commonly evaluated in eco-audits?

- A. Oil equivalent
- B. Solar and wind mix
- C. Hydropower exclusively
- D. Nuclear only

Answer: A

20. In an eco-audit summary chart, which phase often shows the largest energy use for electric appliances?

- A. Transport
- B. Packaging
- C. Use phase

D. Material phase

Answer: C